Reachability, Communication, and Irreducibility

Monte Carlo Statistical Methods

Sai Sampath Kedari
sampath@umich.edu

Contents

1 Level 1: Reachability, Communication, and Irreducibility

This level describes the structural geometry of Markov chains. It explains how states connect to each other, which determines all long–run properties such as recurrence, transience, periodicity, stationary distributions, and ergodicity.

Throughout this section we state the important consequences but postpone all proofs to later levels.

1.1 1. Reachability

1.1.1 1.1 Discrete State Space (Finite or Countable)

For a Markov chain with transition matrix P, we say state j is *reachable* from state i if

$$\exists n \geq 1 \quad \text{such that} \quad (P^n)_{ij} > 0.$$

Intuition. Think of the chain as a directed graph with an edge $i \to j$ whenever $p_{ij} > 0$. Reachability means there is at least one path from i to j with positive probability.

1.1.2 1.2 Continuous or Uncountable State Space

Let $T(x, A)$ be the transition kernel and let $T(x, y)$ be a density if it exists. State y is reachable from x if

$$\exists n \geq 1 \quad \text{such that} \quad T^n(x, y) > 0.$$

Intuition. If the distribution of X_n starting from x assigns positive density near y, then it is possible to reach y in finitely many steps.

2 2. Communication

Two states i and j *communicate* if

$$i \to j \quad \text{and} \quad j \to i.$$

This relation is an equivalence relation: reflexive, symmetric, and transitive. Hence the state space is partitioned into communication classes.

3 3. Communication Classes

A set C is a *communication class* if

$$\forall\, i, j \in C, \quad i \leftrightarrow j.$$

Intuition. Think of a communication class as an island. Inside the island all states can reach each other. Different islands may not communicate at all or may have one–way links.

3.1 4. Closed Classes

A class C is *closed* if

$$i \in C,\ i \to j \implies j \in C.$$

Intuition. A closed class cannot be exited. All recurrent states must live inside closed classes (proved later).

4 5. Irreducibility

A Markov chain is *irreducible* if the entire state space is a single communication class:

$$\forall i, j \in S, \quad i \leftrightarrow j.$$

Intuition. The state space is fully connected. There are no separate islands and no isolated regions.

Irreducibility is a structural property. It does not by itself imply recurrence, aperiodicity, or convergence. Those properties are introduced in later levels.

5 6. Class Properties

A *class property* is a property that is shared by all states in the same communication class.

Examples. Later we prove that the following are class properties:

- recurrence or transience,
- positive or null recurrence,
- periodicity,
- positivity of stationary distribution entries,
- ergodic convergence properties.

If states communicate, then their long–run behavior is linked and cannot differ.

6 7. Consequences of Irreducibility

Here we state the important consequences of irreducibility. Proofs appear in Levels 3 through 7.

6.1 7.1 Finite State Space + Irreducibility

For a finite irreducible Markov chain:

1. All states have the same period (period is a class property).
2. Either all states are recurrent or all are transient.
3. Transience is impossible in finite irreducible chains. Therefore all states are positive recurrent.

4. There exists a unique stationary distribution π.
5. Every entry is strictly positive: $\pi(i) > 0$.
6. If also aperiodic, then ergodicity holds:

$$(P^n)_{ij} \to \pi(j).$$

6.2 7.2 Countable Infinite State Space + Irreducibility

For a countably infinite irreducible chain:

- Period is a class property.
- Recurrence or transience is a class property.
- Positive and null recurrence are meaningful distinctions.
- A stationary distribution exists if and only if the chain is positive recurrent.
- The stationary distribution, if it exists, is unique.
- If the chain is aperiodic and positive recurrent, then

$$(P^n)_{ij} \to \pi(j).$$

6.3 7.3 Uncountable State Space + Irreducibility

For a chain on a continuous state space with kernel $T(x, A)$:

- Irreducibility means that for every measurable set A with positive Lebesgue measure and every x, there exists n such that $T^n(x, A) > 0$.
- Existence of a stationary distribution requires additional regularity conditions such as positive Harris recurrence, minorization, and drift conditions.
- If a stationary density $\pi(x)$ exists and the chain is irreducible and positive Harris recurrent, then π is unique.
- If also aperiodic in the continuous sense, then

$$T^n(x, \cdot) \to \pi(\cdot).$$

These facts form the theoretical foundation of continuous state MCMC algorithms such as Metropolis–Hastings and Gibbs sampling.

7 8. Examples

7.1 Example 1: Irreducible 2 by 2 Chain

$$P = \begin{pmatrix} 0.2 & 0.8 \\ 0.1 & 0.9 \end{pmatrix}$$

Every state can reach every other. This chain is irreducible.

7.2 Example 2: Two Communication Classes

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0.5 & 0.5 \end{pmatrix}$$

The state $\{1\}$ forms a closed class. States $\{2, 3\}$ form another class but cannot reach state 1.

7.3 Example 3: Irreducible but Periodic

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Every state reaches every other, so the chain is irreducible. All states have period 3, so the chain is periodic. A stationary distribution exists, but $(P^n)_{ij}$ does not converge.

8 9. Summary

Level 1 establishes the structural geometry of Markov chains:

- reachability,
- communication,
- communication classes,
- closed classes,
- irreducibility,
- class properties,
- consequences for recurrence, periodicity, and stationary distributions.

All deeper results, such as recurrence theorems, return time formulas, stationary distribution proofs, and ergodicity, will be developed in later levels.